

INFT-1205 – Data Communications & Networking II

Capstone Project

Floor 2 Switch Configuration

Institution: Durham College

By: Group 1

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Document Overview

This document serves as a brief documentation of the FL2 Switch. It includes an overview of the FL2 Switch, along with its configuration.

Device Overview

The FL2 switch is the main switch on Floor 2 of Datacore Solutions. It hosts VLANs for the HR, Management, and System Admins, Accounting, and Legal departments. It connects to FL1 via a 3-lane EtherChannel, configured using LACP. FL2 is fully configured with layer 2 security and includes a Parking Lot VLAN for unused ports.

Configuration

Building configuration...

Current configuration : 11731 bytes

!

! Last configuration change at 08:54:40 UTC Fri Mar 27 2026

!

version 17.6

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption

platform punt-keepalive disable-kernel-core

!

hostname FL2

!

!

vrf definition Mgmt-vrf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

enable secret 9 \$9\$v7e4CT8osQ9cc.\$6FwGg/IH8N48ndPREczEPboEbH9cPo/laC.5lozvauo

!

no aaa new-model

switch 1 provision c9300l-24p-4g

!

!

!

!

!

!

!

!

!

no ip domain lookup

!

!

!

ip dhcp snooping vlan 10,20,30,40,50,60,70,80

ip dhcp snooping

ip arp inspection vlan 10,20,30,40,50,60,70,80

login on-success log

!

!

!

!

!

!

!

!

!

crypto pki trustpoint TP-self-signed-1832108957

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-1832108957

revocation-check none

rsakeypair TP-self-signed-1832108957

!

!

crypto pki certificate chain TP-self-signed-1832108957

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 31383332 31303839 3537301E 170D3236 30333236 31313336
30375A17 0D333630 33323531 31333630 375A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 38333231
30383935 37308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 0100B760 7E92B1EE C157BCA2 82C9DC45 B7021B8F E74A270A 9BCDF74D
EC95D5CD 15BEAE22 341D5DE8 BBBFFE8F 7C7C869C B19FA0C2 BE43B918 38126B4E
460B4B10 EB58AC6E 10FFF4B7 88587921 AEE3BC6A 8DF79825 0B42DA1F 8A43AF31
0C848FBC 81ACE46C FB160F78 A4D46772 5E574CA4 99E30B64 E6935728 91A05E51
1C89D7BA 8778B01E 31BB0BB5 19919A67 E1B12C54 E9FDCB76 DEBF2905 853FA645
66C613C1 02553527 16A534B2 96CF84AF 81D72371 205C4110 473FEADD 434B9B41
32E7F3DA 938A50FD 4A47F205 AC0272E4 D28C28F4 7E3ED0D8 F0C74995 4A50012A
EDD36602 47F378DB 9077026E 9D397668 2CE8689C 1F0A0AE5 1E4F3977 74CD7868

8C141601 E5110203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 148711A7 81E7FB96 94DD0534 9E0B32BE 045F0657
CF301D06 03551D0E 04160414 8711A781 E7FB9694 DD05349E 0B32BE04 5F0657CF
300D0609 2A864886 F70D0101 05050003 82010100 A543E8D8 8F00FF95 90FAEEFC
BEAB8916 7B6E1B26 87DF768A AF1D0C33 70AF35C7 27537582 466B1A83 8434DFC1
F4E7DA87 59181A70 3F864E66 142BEBE6 2E41BB7A 1FF0D5F8 AF706A0B F5CF6597
03E22F59 E866FF23 E88F550D 25F670E3 D54F2B20 EDEF7CF7E FAF180C6 8025B5C7
42B7B059 AEDA40CB D1FF7029 AA11EFBD 647D8639 94E549A3 80A34F4B B9634DAF
A4A9B7CF FFD753A3 09E26202 EEF0A0FB 61C93A30 938A286D 0AF8D719 32D8A45D
50F2E95E A9AE28D0 FE85D138 C06B52C8 1D3754EE D5AA2EFD 8EBFB0F9 348063BF
BB112232 21280F0C 9329CCCE CE4E027B B1B10657 5A0C8357 B2D5FBFD B110E012
AEF5E4B6 35D468F6 B4D336FC 429A2455 CFC3F351

quit

!

!

license boot level network-advantage addon dna-advantage

!

!

diagnostic bootup level minimal

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

memory free low-watermark processor 131688

!

!

redundancy

mode sso

!

!

!

!

crypto engine compliance shield disable

!

!

transceiver type all

monitoring

!

!

class-map match-any system-cpp-police-ewlc-control

description EWLC Control

class-map match-any system-cpp-police-topology-control

description Topology control

class-map match-any system-cpp-police-sw-forward

description Sw forwarding, L2 LVX data packets, LOGGING, Transit Traffic

class-map match-any system-cpp-default

description EWLC Data, Inter FED Traffic

class-map match-any system-cpp-police-sys-data

description Openflow, Exception, EGR Exception, NFL Sampled Data, RPF Failed

class-map match-any system-cpp-police-punt-webauth

description Punt Webauth

class-map match-any system-cpp-police-l2lvx-control

description L2 LVX control packets

class-map match-any system-cpp-police-forus

description Forus Address resolution and Forus traffic

class-map match-any system-cpp-police-multicast-end-station

description MCAST END STATION

class-map match-any system-cpp-police-high-rate-app

description High Rate Applications

class-map match-any system-cpp-police-multicast

description MCAST Data

class-map match-any system-cpp-police-l2-control

description L2 control

class-map match-any system-cpp-police-dot1x-auth

description DOT1X Auth

class-map match-any system-cpp-police-data

description ICMP redirect, ICMP_GEN and BROADCAST

class-map match-any system-cpp-police-stackwise-virt-control

description Stackwise Virtual OOB

class-map match-any non-client-nrt-class

class-map match-any system-cpp-police-routing-control

description Routing control and Low Latency

class-map match-any system-cpp-police-protocol-snooping

description Protocol snooping

class-map match-any system-cpp-police-dhcp-snooping

description DHCP snooping

class-map match-any system-cpp-police-ios-routing

description L2 control, Topology control, Routing control, Low Latency

class-map match-any system-cpp-police-system-critical

description System Critical and Gold Pkt

class-map match-any system-cpp-police-ios-feature

description

ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTData,Transit,D
OT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusARP,McastEndStn,Openflow,Exc
eption,EGRExcption,NflSampled,RpfFailed

!

policy-map system-cpp-policy

!

!

!

!

!

!

!

!

!

!

!

!

interface Port-channel1

description Link to FL1

switchport trunk native vlan 1000

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000

switchport mode trunk

switchport nonegotiate

ip arp inspection trust

```
ip dhcp snooping trust
```

```
!
```

```
interface GigabitEthernet0/0
```

```
vrf forwarding Mgmt-vrf
```

```
no ip address
```

```
negotiation auto
```

```
!
```

```
interface GigabitEthernet1/0/1
```

```
description Link to Host 3
```

```
switchport access vlan 40
```

```
switchport mode access
```

```
switchport port-security maximum 2
```

```
switchport port-security violation restrict
```

```
switchport port-security mac-address sticky
```

```
switchport port-security mac-address sticky 24fb.e389.6c43
```

```
switchport port-security mac-address sticky c84d.4427.37d2
```

```
switchport port-security
```

```
spanning-tree portfast
```

```
spanning-tree bpduguard enable
```

```
ip dhcp snooping limit rate 10
```

```
!
```

```
interface GigabitEthernet1/0/2
```

```
switchport access vlan 999
```

```
switchport mode access
```

```
switchport port-security
```

```
ip arp inspection limit rate 1
```

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/3

description Link to Host 4

switchport access vlan 50

switchport mode access

switchport port-security maximum 2

switchport port-security violation restrict

switchport port-security mac-address sticky

switchport port-security

spanning-tree portfast

spanning-tree bpduguard enable

ip dhcp snooping limit rate 10

!

interface GigabitEthernet1/0/4

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/5

```
switchport mode trunk
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000
switchport nonegotiate
ip arp inspection trust
channel-group 1 mode active
ip dhcp snooping trust
!
```

```
interface GigabitEthernet1/0/6
switchport mode trunk
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000
switchport nonegotiate
ip arp inspection trust
channel-group 1 mode active
ip dhcp snooping trust
!
```

```
interface GigabitEthernet1/0/7
switchport mode trunk
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,1000
switchport nonegotiate
ip arp inspection trust
channel-group 1 mode active
ip dhcp snooping trust
!
```

```
interface GigabitEthernet1/0/8
```

```
switchport access vlan 999
```

```
switchport mode access
```

```
switchport port-security
```

```
ip arp inspection limit rate 1
```

```
shutdown
```

```
spanning-tree bpduguard enable
```

```
ip dhcp snooping limit rate 1
```

```
!
```

```
interface GigabitEthernet1/0/9
```

```
switchport access vlan 999
```

```
switchport mode access
```

```
switchport port-security
```

```
ip arp inspection limit rate 1
```

```
shutdown
```

```
spanning-tree bpduguard enable
```

```
ip dhcp snooping limit rate 1
```

```
!
```

```
interface GigabitEthernet1/0/10
```

```
switchport access vlan 999
```

```
switchport mode access
```

```
switchport port-security
```

```
ip arp inspection limit rate 1
```

```
shutdown
```

```
spanning-tree bpduguard enable
```

```
ip dhcp snooping limit rate 1
```

!

interface GigabitEthernet1/0/11

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/12

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/13

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/14

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/15

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/16

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/17

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/18

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/19

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/20

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/21

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/22

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/23

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/0/24

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/1/1

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/1/2

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/1/3

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface GigabitEthernet1/1/4

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface AppGigabitEthernet1/0/1

switchport access vlan 999

switchport mode access

switchport port-security

ip arp inspection limit rate 1

shutdown

spanning-tree bpduguard enable

ip dhcp snooping limit rate 1

!

interface Vlan1

no ip address

!

ip forward-protocol nd

ip http server

ip http secure-server

!

!

!

!

!

!

control-plane

service-policy input system-cpp-policy

!

banner motd ^CC Unauthorized Access is Prohibited ^C

!

line con 0

password 7 15562C1E0B3F3B750B1C1B

login

stopbits 1

line vty 0 4

password 7 01572116541E165E177877

login

transport input ssh

line vty 5 31

login

transport input ssh

!

!

!

!

!

!

!

end